ATHENA UNIVERSITY - DEPARTMENT OF COMPUTER SCIENCE

CS101: Introduction to Artificial Intelligence (Fall 2026)

Instructor: Prof. Ada Lovelace | Email: lovelace@athena.edu

Office: Turing Hall, Room 302 | Office Hours: Tuesdays & Thursdays 2:00 - 4:00 PM

Course Description: This course covers foundations of state-space search, heuristics, vector embeddings, neural representations, and Retrieval-Augmented Generation (RAG).

GRADING POLICY AND ASSESSMENT BREAKDOWN:

- Final Capstone Project: 40% (Due December 15th)
- Midterm Examination: 30% (In-class on October 24th)
- Weekly Programming Labs: 20%
- Seminar Participation & Quizzes: 10%

Late Submission Policy: Assignments submitted after the deadline incur a 10% deduction per day up to a maximum of 3 days. Submissions beyond 3 days receive zero credit.

WEEKLY LECTURE SCHEDULE:

Week 1: History of AI, Agents, and Search Algorithms (A* and Minimax)

Week 2: Vector Embeddings and Dense Representations (Cosine Similarity & FAISS)

Week 3: Transformer Architectures and Self-Attention Mechanisms

Week 4: Retrieval-Augmented Generation (RAG) Architecture and Provenance

Week 5: Multi-Agent Systems: Supervisors, Specialists, and Tool Calling

ACADEMIC HONESTY & CODE OF CONDUCT:

All work must represent the student's own effort. Collaboration on conceptual ideas is encouraged, but sharing code or lab solutions constitutes academic misconduct.

Required Textbook: Artificial Intelligence: A Modern Approach (Russell & Norvig, 4th ed).